

VAE vs. DDPM

for Image Generation on Fashion-MNIST

A controlled comparison of distribution quality, diversity, and computational efficiency

RESEARCH QUESTION

How much generation quality does iterative diffusion gain over latent-variable decoding?

What computational cost accompanies it?

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METHODS AND CONTROLS

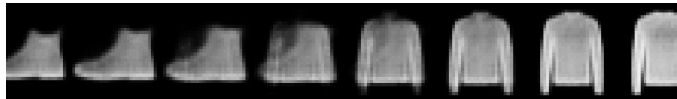
Same evaluation pipeline; different generation mechanisms.

Controlled setup

Fashion-MNIST 60k/10k · 28×28 grayscale · 5,000 generated samples/model · CNN evaluator 91.34% accuracy · average of 10 seeds

VAE

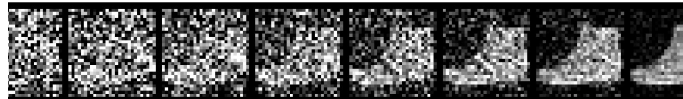
Latent-variable decoding



Encoder $q(z|x)$ learns a 32-D latent space
Generate: sample $z \sim N(0, I)$,
then one **decoder** pass
1.70M parameters · 25 epochs

DDPM

Iterative reverse denoising



Noise-prediction U-Net with cosine schedule
Generate: 100 sequential reverse-denoising steps
5.22M parameters · 35 epochs

Quality: FID/KID + confidence **Diversity:** coverage + entropy **Cost:** time + latency + memory

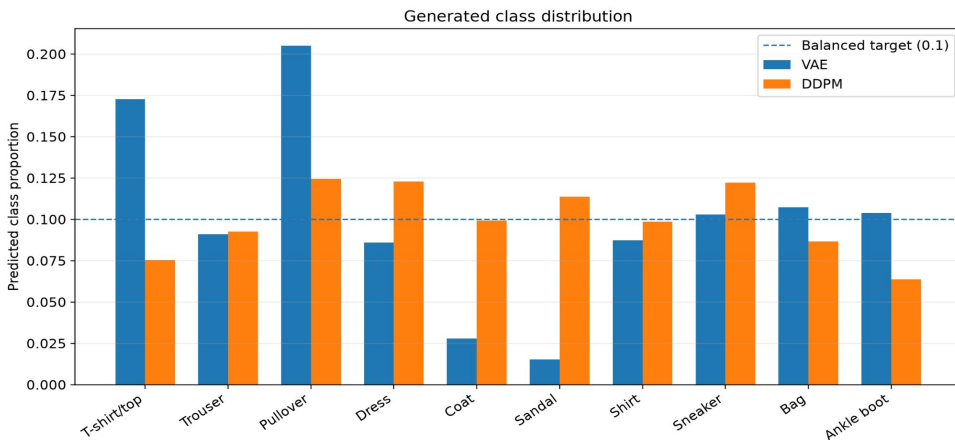
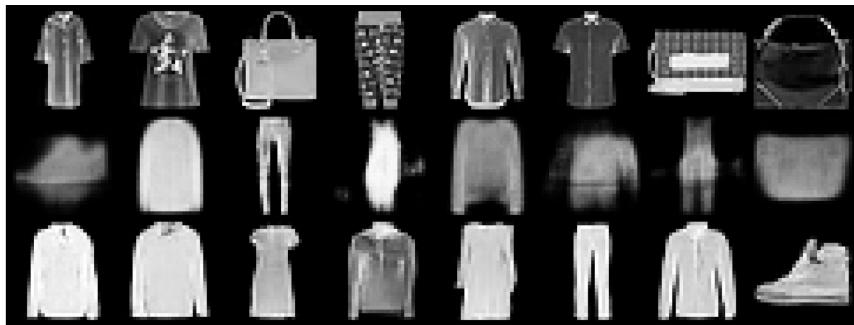
Qualification: controlled system comparison, not a capacity-matched causal ablation.

DDPM generated more recognizable samples with better feature matching and class balance.

Real

VAE

DDPM



QUANTITATIVE EVIDENCE

Feature Quality

FID ↓ KID ↓
51.1 vs. **7.7** **1.87** vs. **0.33**
(Confirms FID)

Recognizability

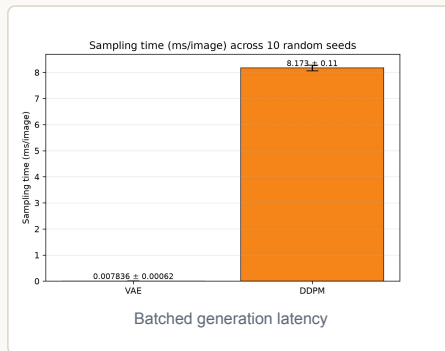
41% vs. **66%**
Confidence ≥ 0.8 ↑

Class Balance

0.931 vs. **0.991**
Entropy ↑

Both models cover all 10 classes, but the VAE overproduces several categories.

DDPM's stronger average quality came with substantially greater compute.



Resource multiplier: DDPM / VAE

11.7× training time

~1,043× Observed batched sampling time per image

11.2× peak GPU memory

3.1× parameters

Why Batched sampling time per image explodes: parameter count rises only 3.1×, but DDPM generation evaluates the U-Net 100 times. The VAE uses one decoder pass.

VAE strength: fast, compact, smooth latent interpolation
Limitation: blurrier samples and weaker feature match

DDPM strength: stronger FID/KID, confidence, sharpness, balance
Limitation: sequential sampling and higher memory cost

Mean ± sample SD across ten runs on one RTX 5070 Ti. Sampling uses VAE batch 512 and DDPM batch 1,024; values are amortized throughput, not batch-one latency. Training schedules were 25 and 35 epochs.

Neither model dominates: diffusion buys quality; VAE buys speed.

Choose DDPM when the priority is...

- Distribution match and recognizability
- Balanced predicted class coverage
- Fidelity matters more than inference cost

Evidence: lower FID/KID, higher confidence, entropy 0.991 vs. 0.931

Choose VAE when the priority is...

- Low-latency generation
- Lower memory/training budget
- Direct 32-D latent interpolation

Evidence: ~1,043× lower ms/image; 11.7× less training time; 11.2× less peak allocation

Validity limits

Unequal number of parameters; no human study; Fashion-MNIST only;
no formal significance tests; 100-step vanilla DDPM

Next experiments

matched parameter/compute budget · 10/25/50/100 DDPM-step
Pareto curve · per-class diversity

Takeaway: Across ten Fashion-MNIST runs, our 100-step DDPM consistently favored classifier-based quality metrics, while our VAE favored computational efficiency.